

Muzammal Hussain

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● ABOUT ME

Full-Stack Developer with 3+ years of experience building scalable SaaS platforms, AI-powered applications, and cloud-based systems using **Node.js, NestJS, React.js, PostgreSQL, and TypeScript**. I've worked across the full product lifecycle, from frontend interfaces and backend APIs to **AI integrations, GIS workflows, cloud deployments, and production debugging**. My experience includes building **multi-tenant platforms, RBAC systems, payment integrations, regulatory-intelligence workflows, and LLM/RAG pipelines** using OpenAI and LangChain. I also have experience working with geospatial systems using **ArcGIS and PostGIS**, building planning and automation workflows that combine structured backend systems with AI-driven analysis and document processing. I enjoy building clean, maintainable systems, solving complex backend challenges, and collaborating with teams to turn complex product ideas into reliable real-world solutions.

● WORK EXPERIENCE

SOFTWARE ENGINEER – FUTURENOSTICS – 31/10/2024 – Current – E11, ISLAMABAD, PAKISTAN

- Contributed across the full **AEDI platform**, working on **React frontend applications, NestJS APIs, FastAPI AI services, GIS integrations, billing systems, and cloud deployments** for a multi-tenant municipal planning SaaS product focused on automated zoning and land-development assessments.
- Developed municipality-specific **GIS and AI workflows** using **ArcGIS, PostGIS, FastAPI, LangChain, OpenAI, and RAG pipelines** to process zoning bylaws, parcel data, permitted-use analysis, dimensional checks, and planning constraints across multiple Canadian municipalities.
- Built and maintained scalable **backend and cloud infrastructure** using **NestJS, PostgreSQL, Redis, Docker, AWS ECS, and GCP**, while also contributing to **Stripe billing workflows, HIPAA-compliant systems, RBAC implementations, automation services, and production debugging** across distributed systems.
- Built a **Medicaid compliance platform** on **Google Cloud Platform (GCP)** using **Cloud Run, Artifact Registry, and Docker-based deployments**.
- Developed a **SaaS-based Sales Management System** for e-commerce businesses to manage **sales, inventory, and order workflows**.
- Implemented **Role-Based Access Control (RBAC)** systems with granular roles and permissions to improve **application security and access management**.
- Created a **HIPAA-compliant dynamic form builder** for a **Case Management Hub**, enabling organizations to create, distribute, and manage secure healthcare workflows.
- Automated operational workflows by integrating **Monday.com with Google Sheets**, reducing manual data handling and improving process efficiency.
- Collaborated with **cross-functional teams in Agile environments**, participating in planning, task estimation, and feature delivery using **Jira**.

BACKEND DEVELOPER – ARGONTEQ – 31/05/2023 – 30/11/2024 – MODEL TOWN, LAHORE

- Developed **backend services** for an **educational CRM platform**, managing student profiles, enrollments, authentication workflows, and academic progress tracking for admins and parents.
- Built secure **authentication and user management modules** with **Role-Based Access Control (RBAC)** to support different access levels across platform users.
- Developed **Willo Box**, an **AI-powered document query platform** using **OpenAI APIs and LangChain**, enabling users to upload PDFs and retrieve context-aware answers through semantic search workflows.

- Contributed to the backend development of a **crypto exchange platform** using **Node.js** and **MongoDB**, handling user management, transaction workflows, and secure data operations.
- Built **backend systems** for a gaming platform with support for **game management, tournament organization, and admin-controlled content publishing workflows**.
- Integrated **Google Reviews** into a restaurant tipping platform to improve customer engagement and restaurant interaction experiences.
- Developed backend workflows for a **pharmacy marketplace platform**, enabling pharmacies to manage registrations, medication listings, and transaction processing between buyers and sellers.
- Collaborated with **frontend developers, QA teams, and project managers** in **Agile environments** to deliver scalable backend features and API integrations.

● **SKILLS**

Jest, Vitest, Playwright, Postman, ESLint, Prettier, Husky, Storybook

Frontend

React.js | TypeScript | Vite | Bootstrap | Tailwind CSS | Zustand | TanStack Router/Query | Shadcn/UI

Backend

JavaScript | Node.js | NestJS | Express.js | GraphQL | FastAPI | RabbitMQ | WebSockets |
Microservices Architecture

Databases

MongoDB | PostgreSQL | MySQL | Redis | Vectore Database | Firestore

Version Control & Collaboration

Git | GitHub | GitLab | GitHub Actions | Husky | Jira | ClickUp | Slack

Cloud & DevOps

Docker | AWS (EC2, ECS, S3) | Google Cloud Platform (Cloud Run, Artifact Registry, GCS)

Testing

Jest | ESLint | Prettier | Storybook | Unit Testing | Integration Testing | Postman Automation Testing |
Playwright

AI & Automation

LangChain | RAG Pipeline | Semantic Search | PDF Processing | OpenAI

● **PROJECTS**

AEDI — Municipal Planning & Zoning SaaS

Write here the Multi-tenant SaaS platform for Canadian land development and municipal planning, featuring automated zoning assessments, parcel analysis, GIS integrations, and AI-powered planning workflows.

Built across React, NestJS, FastAPI, PostGIS, ArcGIS, OpenAI, LangChain, and AWS ECS, including RAG pipelines, geospatial workflows, billing systems, and municipality-specific regulatory intelligence.

Case Management Hub (CMH)

HIPAA-compliant case management platform with dynamic form-building capabilities, secure workflows, and role-based access management.

Developed secure backend services, RBAC workflows, and configurable form management systems for healthcare-related operational processes.

Sales Management System

SaaS-based sales and inventory management platform for e-commerce businesses to manage products, orders, inventory tracking, and operational workflows.
Implemented backend APIs, authentication workflows, and role-based permission systems for secure business operations.

Willo Box — AI Document Query Platform

AI-powered document analysis platform that allows users to upload PDFs and retrieve context-aware answers using semantic search and LLM-based retrieval workflows.
Built using OpenAI APIs, LangChain, Node.js, and vector-based document processing pipelines.

Monday.com ↔ Google Sheets Automation

Workflow automation service for synchronizing operational data between Monday.com boards and Google Sheets.
Built automation pipelines to reduce manual data handling and streamline reporting and operational processes.

Katanalnu — Gaming Platform

Gaming platform with support for game publishing, tournament organization, and user interaction features, allowing admins to manage content and players to access and participate in games.
Built backend workflows for game management, tournament systems, and admin-controlled content publishing using scalable API-driven architecture.

Obon — Restaurant Tipping Platform

Restaurant tipping and customer engagement platform designed to enhance customer interaction and feedback collection for restaurants.
Integrated **Google Reviews APIs** and developed backend workflows to improve restaurant visibility, customer engagement, and feedback-driven experiences.

● **EDUCATION AND TRAINING**

07/10/2021 – 31/05/2025 Lahore
BS SOFTWARE ENGINEERING Virtual University of Pakistan

Final grade 3.69/4

22/05/2014 – 05/04/2016
F.SC PRE ENGINEERING Govt. Degree College

Final grade 82%